5A · 6A · 10A · Atmosphere · EvolLand

Test 3 Master Test Review

Simplified · 38 Review Questions

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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REVIEW

Test 3 Master Test Review: Simplified

38 short questions to focus on as you prepare for Test 3, organized by subject. Each one tells you where to find the answer. Most can be answered in a word or a sentence; write your answers on the lines.

WHERE TO FIND THE ANSWERS	PASSWORD
Module 5A	Module5A
5AE1	5AE1SG
5AE2	5AE2SG
Atmospheric Chart	Atm
Atmospheric Pressure exercise	AtmosSG
Ancient Atmospheres video	AASG
EvolLand exercise	EvolSG
Module 6A	Module6A
6AE1	6AE1SG
Module 10A	Module10A

The Earth's Atmosphere

1. Excluding water vapor, what four gases make up most of the atmosphere?

Find answers in: Module 5A, 5AE1

2. Name the four layers of the atmosphere, from the ground up.

Find answers in: Module 5A, 5AE1, Atmospheric Chart

3. In which two layers does the temperature DECREASE as you go higher, and in which two does it INCREASE?

Find answers in: Module 5A, 5AE1, Atmospheric Chart

4. Which two gases in the stratosphere protect life from deadly UV radiation?

Find answers in: Module 5A, 5AE1

5. Up to what frequency does the ionosphere reflect radio waves?

Find answers in: Module 5A, 5AE1

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6. What sphere traps the charged particles of the solar wind?
Find answers in: Module 5A, 5AE1

7. Name the six main ingredients of Earth-based weather.
Find answers in: Module 5A, 5AE2

8. Name the four forms of precipitation.
Find answers in: Module 5A, 5AE2

9. Are fog, dew, and clouds forms of precipitation?
Find answers in: Module 5A, 5AE2

10. In summer, what are cumulonimbus (storm) clouds mostly made of?
Find answers in: Module 5A, 5AE2

11. Name three greenhouse gases.
Find answers in: Module 5A, 5AE2

12. What kind of weather does high pressure bring, and what kind does low pressure bring?
Find answers in: Module 5A, 5AE2

Atmospheric Pressure Exercise

13. How does atmospheric pressure change as you go higher?
Find answers in: Atmospheric Pressure exercise

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14. What is one Atmosphere in millibars, psi, and inches of mercury?

Find answers in: Atmospheric Pressure exercise

Ancient Atmospheres

15. Which organisms caused the Great Oxygenation Event, and about when?

Find answers in: Ancient Atmospheres video

EvolLand Exercise

16. What was the Cambrian Explosion, and about when did it happen?

Find answers in: EvolLand exercise

17. During the Carboniferous Period, about 360 to 300 million years ago, oxygen climbed to about what level, and what happened to insects?

Find answers in: EvolLand exercise

18. About how long ago did our species, Homo sapiens, evolve?

Find answers in: EvolLand exercise

The Nature of Light

19. What is optics?

Find answers in: Module 6A, 6AE1

20. What is the simple definition of light?

Find answers in: Module 6A, 6AE1

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21. Rounded, what is the speed of light in a vacuum, in miles/s and in km/s?

Find answers in: Module 6A, 6AE1

22. Who first used astronomy to determine the speed of light?

Find answers in: Module 6A

23. What do temporally coherent and spatially coherent mean? Which is laser light?

Find answers in: Module 6A, 6AE1

24. List the electromagnetic spectrum in order of increasing frequency. What wavelengths can humans see?

Find answers in: Module 6A, 6AE1

25. What are the three ways light interacts with matter?

Find answers in: Module 6A, 6AE1

26. Name the five classical phenomena of light.

Find answers in: Module 6A, 6AE1

27. What is the difference between specular and diffuse reflection?

Find answers in: Module 6A

The Exploration of Space

28. What was Sputnik 1?

Find answers in: Module 10A

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29. What was the first satellite the United States launched successfully?
Find answers in: Module 10A

30. Who was the first person to orbit the Earth?
Find answers in: Module 10A

31. Who were the first and second Americans in space?
Find answers in: Module 10A

32. Who was the first American to orbit the Earth?
Find answers in: Module 10A

33. How many men have walked on the Moon, and where did Apollo 11 land?
Find answers in: Module 10A

34. Who were the first man and the first woman to fly solo nonstop across the Atlantic?
Find answers in: Module 10A

35. Which mission sent back the first close-up images of Jupiter?
Find answers in: Module 10A

36. Which mission sent back the first close-up images of Saturn?
Find answers in: Module 10A

37. What was the key first of the Viking missions?
Find answers in: Module 10A

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38. Which spacecraft is the only one to have visited Uranus and Neptune?

Find answers in: Module 10A

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